

# Wirelessly Transmitting A High-Frequency Corrupted Audio File with BPSK Modulation Using USRP

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## I. INTRODUCTION

The aim of this project is to build a simple wireless speech communication system by combining DSP and telecommunication techniques. In a normal speech transmission system, one possible method would be to send the whole speech waveform directly. However, this is not very efficient because a speech signal contains many samples, and transmitting all of them requires a high data rate. For this reason, in this project we try to represent the speech signal with fewer parameters and transmit these parameters instead of the original waveform.

The given input signal is a corrupted speech signal sampled at 48 kHz. Before using it in the communication system, it has to be cleaned and converted to an 8 kHz speech signal. After this preparation step, the speech is divided into frames and analyzed using LPC. The LPC coefficients are converted to LSF parameters because they are more suitable for quantization. In addition to LSF coefficients, gain, pitch, and voiced/unvoiced information are also extracted for each frame. These values are then quantized and converted into bits.

After the DSP part, the binary data is transmitted using BPSK modulation with USRP. At the receiver side, the received signal is demodulated, the bitstream is recovered, and the speech parameters are reconstructed. Finally, an approximate speech signal is synthesized from these recovered parameters. Since the transmitted data goes through a real wireless channel, the received bits may contain errors. Therefore, the quality of the reconstructed speech depends on both the DSP representation and the communication link performance.

The motivation of this project is to see how speech can be transmitted more efficiently by using a parameter-based representation instead of sending raw samples. This makes the project useful because it shows both sides of a practical communication system: reducing the amount of data with DSP and sending that data over a wireless link with digital modulation. In this way, the project gives a clearer understanding of how speech coding and wireless communication can work together

in an end-to-end system.

## II. PIPELINE

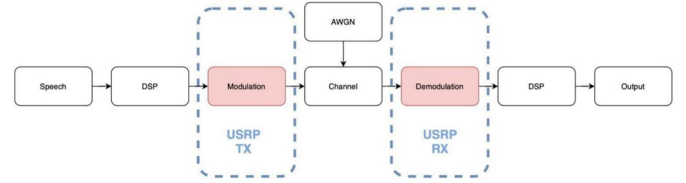


Fig. 1: Pipeline of the Project

Figure 1 shows the general pipeline of our project. The system starts with a corrupted speech signal and ends with an approximately reconstructed speech signal at the receiver side. The main idea is not to transmit the original speech waveform directly. Instead, the speech signal is first processed with DSP methods and represented by a smaller set of parameters. These parameters are then converted into bits, transmitted wirelessly, recovered at the receiver, and used to synthesize the speech again.

At the beginning of the system, the input speech signal is prepared for further processing. The original signal is sampled at 48 kHz and contains unwanted high-frequency components. Because of this, the signal is first passed through a low-pass filter. After filtering, it is downsampled to 8 kHz. This prepared 8 kHz signal is used as the speech input for the DSP part of the system.

In the first DSP block, the speech signal is divided into short frames. In our implementation, each frame is analyzed using LPC. The LPC coefficients are converted into LSF parameters. LSF parameters are used because they are more suitable for quantization and reconstruction. For each frame, gain, pitch, and voiced/unvoiced information are also extracted. These parameters describe the main properties of the speech frame without sending all speech samples directly.

After the speech parameters are extracted, they are quantized and converted into binary form. Each frame is then

placed into a packet structure. The packet contains a preamble, a frame ID, and the payload bits. The preamble helps the receiver find the beginning of a packet. The frame ID helps the receiver understand which speech frame the received payload belongs to. This makes the receiver side more organized when recovering the transmitted frames.

The next stage is the modulation block. In this project, BPSK modulation is used because it is simple, reliable, and easy to implement with USRP. In BPSK, each bit is mapped to one of two symbol values, so the receiver can decide the transmitted bit by checking the sign of the received symbol. This makes the demodulation stage easier and more robust. Other modulation techniques such as QPSK or QAM could also be used. However, these methods are more sensitive to noise, phase errors, and synchronization problems. Since the main aim of this project is reliable transmission of speech parameters rather than very high data rate, BPSK was a suitable choice. After the BPSK mapping, root raised cosine filtering is applied for pulse shaping. This produces the baseband signal that is sent to the USRP transmitter.

The USRP transmitter sends the signal through the wireless channel. The channel block represents the real transmission environment between the transmitter and receiver. In practice, the transmitted signal can be affected by noise, attenuation, timing errors and phase changes. These effects make the received signal different from the transmitted signal.

At the receiver side, the USRP receives the wireless signal and converts it into a complex baseband signal. This signal is processed in MATLAB. Matched filtering, timing offset selection, phase correction, and hard decision are applied to recover the transmitted bits. After bit recovery, the receiver searches for the preamble and extracts the frame ID and payload bits.

In the final DSP block, the recovered payload bits are converted back into speech parameters. The recovered LSF, gain, pitch, and voiced/unvoiced values are used for speech reconstruction. The LSF values are converted back into LPC filter coefficients, and an excitation signal is generated depending on whether the frame is voiced or unvoiced. Finally, the reconstructed speech signal is obtained at the output.

#### *A. Characteristics of The Audio File*

The audio file used in this project is a corrupted speech signal sampled at 48 kHz. Since the signal is corrupted, it contains unwanted components in addition to the actual speech information. These unwanted components can be seen especially in the frequency spectrum of the signal. Because of this, the audio file cannot be used directly for the speech parameter extraction part.

Another important characteristic of the audio file is its sampling rate. A 48 kHz sampling rate means that the signal has many samples per second. If we tried to transmit this speech signal directly, the required data rate would be very high. For example, assuming 16 bits per sample, transmitting the raw speech signal would require a large number of bits

every second. This is not efficient for a simple wireless communication system.

For this reason, the speech signal is first prepared before transmission. The high-frequency components are removed using a low-pass filter, and the signal is downsampled to 8 kHz. This is enough for the speech processing part because most of the important speech information is located in the lower frequency range. After this step, instead of transmitting the full waveform, the signal is represented using LPC/LSF-based speech parameters.

Using LPC/LSF parameters makes the transmission more efficient. Instead of sending all speech samples, only a limited number of parameters such as LSF coefficients, gain, pitch, and voiced/unvoiced information are transmitted for each frame. Therefore, the audio file is converted from a large raw waveform into a more compact representation before the communication stage.

#### *B. System Requirements*

The system required both software and hardware components. On the software side, MATLAB was used as the main development environment. All signal processing, parameter extraction, bitstream generation, modulation, demodulation, BER calculation, and speech reconstruction parts were implemented in MATLAB.

For the hardware part, the NI2900 USRP/B200 platform type was used as the transmitter and receiver device. For wireless transmission, two SMA703 triband antennas are connected to the Tx and Rx channels. In the earlier telecom progress stage, the system was first tested mainly to verify that the USRP transmission and reception worked correctly. After this stage, antennas were used instead of only relying on a direct or simple setup, so the final experiments were closer to a real wireless communication link. The system also required proper gain settings, because the received signal quality changed significantly depending on the TX and RX gain values.

The project also had signal and processing requirements. The input audio had to be a corrupted speech signal sampled at 48 kHz, and it had to be prepared as an 8 kHz speech signal before the DSP stage. After that, the speech information had to be represented by parameters instead of raw samples. For this reason, the system needed frame-based LPC/LSF analysis, gain estimation, pitch estimation, voiced/unvoiced decision, quantization, and binary packet generation.

For the communication part, the binary data had to be transmitted wirelessly using BPSK through the USRP. The receiver had to recover the bitstream from the received baseband signal, detect packets using the preamble, identify frames using the frame ID, and reconstruct the transmitted speech parameters. Finally, the system had to synthesize an approximate speech signal and evaluate the result using BER and reconstructed speech quality.

#### *C. Preprocessing & Prepping The Audio File For Transmission*

Before extracting speech parameters, the audio signal had to be prepared properly. The original file was sampled at 48

kHz and it contained unwanted high-frequency components. If this signal was used directly, these high-frequency parts could affect the LPC analysis and also increase the amount of data unnecessarily. Because of this, the first step was filtering the signal.

A low-pass FIR filter was used before downsampling. The passband frequency was selected as 3500 Hz and the stopband frequency was selected as 4000 Hz. This choice is suitable for speech because most of the useful speech information is located below this range. The filter removes the higher frequency components before the sampling rate is reduced. This step is important because downsampling without filtering can cause aliasing.

After filtering, the signal was downsampled from 48 kHz to 8 kHz. The 8 kHz sampling rate was enough for the speech processing part of the project and it also reduced the number of samples significantly. This made the later DSP stages easier and reduced the amount of information that had to be represented before transmission.

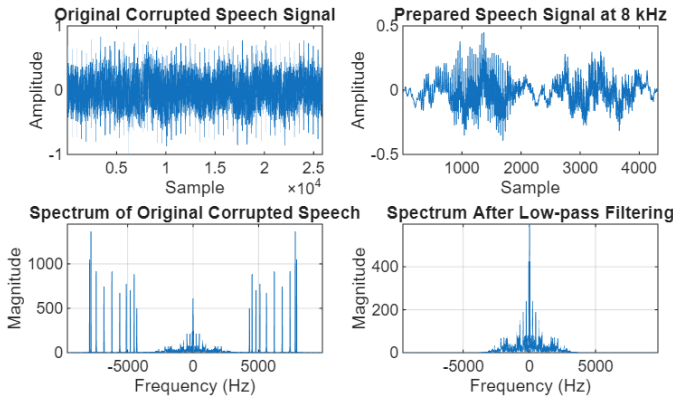


Fig. 2: The speech signal and its frequency spectrum

After obtaining the 8 kHz speech signal, the signal was divided into short frames. In our implementation, each frame length was selected as 160 samples, which corresponds to 20 ms at 8 kHz. This frame length is useful because speech can be assumed to be almost stationary over short time intervals. A 40-sample overlap was also used between frames. This helped the analysis change more smoothly from one frame to the next.

Before applying LPC analysis, a Hamming window was applied to each frame. Windowing was used to reduce sharp edges at the beginning and end of frames. Without windowing, frame boundaries could introduce artificial discontinuities and affect the parameter extraction. After this step, each windowed frame was ready for LPC-based analysis.

For each frame, LPC coefficients were calculated with order 10. Then these coefficients were converted into LSF parameters. Along with LSF values, gain, pitch, and voiced/unvoiced information were also extracted. These parameters were later quantized and converted into bits for transmission.

In our implementation, 10 bits were used for each LSF coefficient, 8 bits were used for gain, 8 bits were used for pitch, and 1 bit was used for the voiced/unvoiced flag. Since each

frame had 10 LSF coefficients, the speech parameter payload size became

$$10 \times 10 + 8 + 8 + 1 = 117$$

bits per frame.

After this step, each speech frame was represented by 117 bits instead of its original audio samples. Therefore, the transmitter only needed to send the extracted speech parameters rather than sending the full speech waveform.

#### D. Transmitting & Receiving The Signal via USRP

After preparing the speech parameters and converting them into bits, the next step was sending these bits with the USRP. In this part, we did not send the speech waveform itself. We sent the bitstream that was created from the quantized LSF, gain, pitch, and voiced/unvoiced parameters.

Before modulation, the bits were arranged as packets. Each packet had a preamble, a frame ID, and payload bits. The payload contained the speech parameters of one frame. The preamble was added because the receiver needed a known bit pattern to find the start of a packet. The frame ID was also added so that the receiver could understand which speech frame the packet belonged to.

For modulation, we used BPSK. In our system, bit 0 was mapped to -1 and bit 1 was mapped to +1. We used BPSK because it is simple and reliable. Since there are only two possible symbol values, the receiver can recover the bit by checking the sign of the received sample. Other modulation methods such as QPSK or QAM could also be used, but they would be more sensitive to noise and phase errors. For this project, reliable bit recovery was more important than sending data at a very high rate, so BPSK was a suitable choice.

After BPSK mapping, we applied a root raised cosine filter. This was used for pulse shaping. The aim was to make the transmitted baseband signal smoother before sending it to the USRP. In our implementation, the samples per symbol value was 8, the roll-off factor was 0.35, and the filter span was 10 symbols.

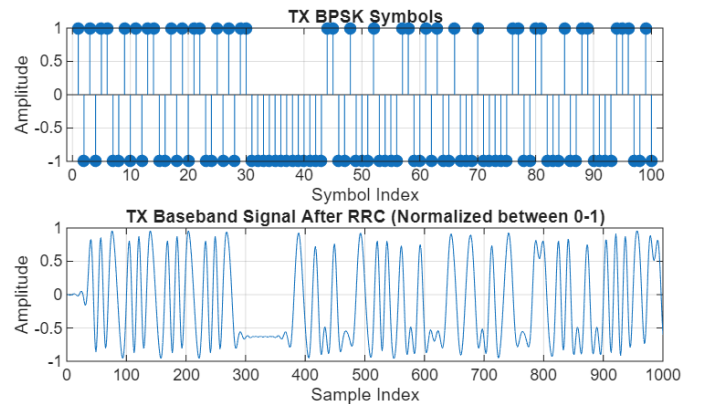


Fig. 3: Baseband signal - Tx channel

**TX Constellation Diagram (BPSK - After Modulation)**

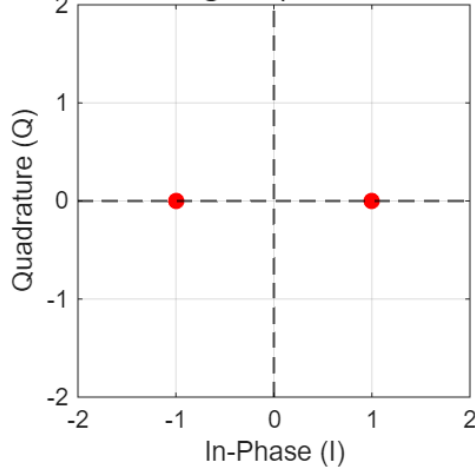


Fig. 4: Constellation diagram of BPSK modulated speech signal

The baseband signal was then transmitted using a USRP. The center frequency was set to 433 MHz. This value was chosen experimentally according to the USRP and antenna setup. The main purpose of the project was not to compare different carrier frequencies, but to transmit the generated BPSK baseband signal over a real wireless link. Therefore, a frequency that worked reliably with the available setup was used. During the experiments, the transmitter and receiver were configured with the same center frequency so that the received signal could be downconverted to baseband correctly.

The master clock rate was 20 MHz and both the interpolation and decimation factors were set to 500. Therefore, the baseband sampling rate of the USRP baseband sampling rate became

$$F_s = \frac{20 \text{ MHz}}{500} = 40 \text{ kHz}.$$

The transmitter and receiver gains were also adjusted experimentally. We observed that gain selection was important. When the gain was too low, the receiver could not recover the bits correctly. Also, unsuitable gain values caused worse received signal quality.

At the receiver side, the USRP collected complex baseband samples. These samples were processed in MATLAB. First, the DC component was removed. Then, the same root raised cosine filter was used as a matched filter. After that, the receiver corrected, searched for the best timing offset, corrected the phase, and made hard decisions to recover the transmitted bits.

After the bitstream was recovered, the receiver searched for the preamble. When a packet was found, the frame ID and payload bits were extracted. If more than one packet was detected for the same frame, the one with the best preamble score was selected. Then the recovered payload bits were used for speech parameter reconstruction.

Finally, the transmission quality was evaluated using BER. The recovered bits were compared with the original transmitted bits. A low BER showed that the USRP transmission

and reception worked correctly. When the BER was low, the recovered speech parameters were close enough to the transmitted parameters, so the reconstructed speech became understandable.

### E. Speech Reconstruction

At this point, the bitstream is received, and the signal is ready to be demodulated. The signal's mean is removed in case of any DC introduction during transmission. A matched filter is applied with Root Raised Cosine to match with the transmission filter and improve SNR as a result. Later time offset search and phase correction are implemented to process the received signal without phase and time shift and achieve higher accuracy results.

The packets were tagged by their frame IDs, so they could be reordered after reception and the payload bits corresponding to a low-level acoustic representation of the speech parameters were reconstructed. The bitstream was firstly separated to LSF, gain, pitch and voiced/unvoiced parameters. As these values were quantized during transmission, they were first dequantized to a value close to their original one. For frames that had failed reconstruction, the parameters used for the previous successfully reconstructed frame were reused where only the pitched and unpitched gains were reduced in value from that used in the last validly reconstructed frame to avoid large jumps or discontinuities in the generated speech.

Once the parameters were estimated, the LSF coefficients had to be converted back to LPC coefficients. For voiced frames, an impulse train excitation signal was produced by using the retrieved pitch value. To replicate silent letters for unvoiced frames, a white Gaussian noise was the choice of excitation source. The gain parameter recovered was then used to scale the excitation signal and run the LPC synthesis filter. As a result, you got a speech frame according to the transmitted parameters.

Finally, all synthesized frames were connected with respect to their frame id to form the reconstructed speech signal. The resulting signal was normalized to remove DC offset and prevent clipping before being played back and saved as an audio file. To evaluate the reconstruction performance, the recovered speech was compared with the original processed speech signal using SNR metrics.

## III. RESULTS & DISCUSSIONS

The system evaluation is done with bit error rate, signal-to-noise ratio, and audio evaluation.

### A. BER, Sound Quality & System Evaluation

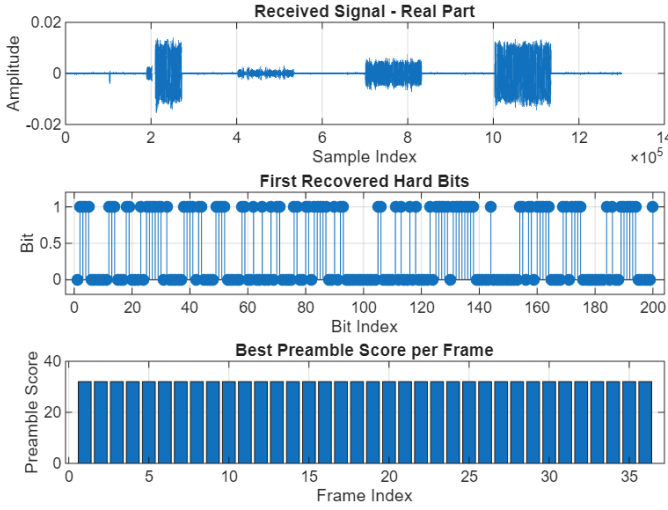


Fig. 5: Received signal evaluation

Figure 5 presents the main results obtained during signal reception and packet detection. In the first graph, the received signal appears as separate bursts because the bitstream was transmitted repeatedly during specific time intervals. The regions between these bursts mainly contain background noise because no useful packet is being transmitted at those moments. The amplitudes of the received bursts are not exactly the same. This is expected in a real wireless system because the received signal strength can change due to path loss, reflections from surrounding objects, multipath propagation, antenna position, and small variations in the USRP transmitter and receiver gains. In addition, the Root Raised Cosine pulse-shaping filter spreads each BPSK symbol over several samples. Therefore, the instantaneous amplitude also changes according to the transmitted bit sequence, even though the original BPSK symbols are represented by only +1 and -1. These amplitude differences do not directly indicate packet loss or bit errors, because bit decisions are made only after matched filtering, synchronization, normalization, and symbol sampling. The second graph shows the recovered hard-decision bits, where the processed symbol values are converted into a clear sequence of zeros and ones. The last graph shows the best preamble correlation score for each of the 36 speech frames. The scores are high and almost equal for all frames, which means that the receiver reliably found the known preamble pattern and correctly detected the beginning of each packet. Therefore, Figure 5 shows that, although the raw received signal contains amplitude variations caused by the practical wireless channel and hardware, the receiver still successfully detected the packets and recovered the transmitted binary data.

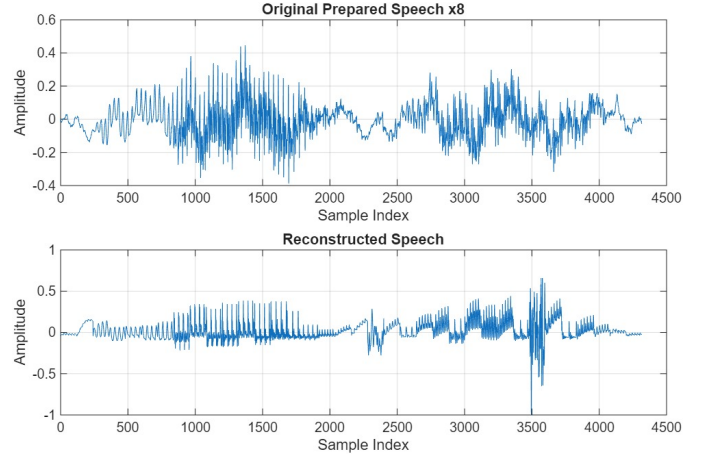


Fig. 6: Speech signals (original and recovered)

Figure 6 compares the prepared original speech signal with the speech reconstructed from the received parameters. The reconstructed waveform is not expected to be exactly the same as the original waveform because the system does not transmit the original audio samples. Instead, each speech frame is represented by ten LSF coefficients, one gain value, one pitch value, and one voiced/unvoiced flag. At the receiver, these parameters are used to generate a new speech signal with an LPC synthesis filter. Therefore, the receiver preserves the general speech information, but it cannot reproduce the exact sample values or the exact phase of the original waveform. For voiced frames, an impulse train is generated according to the recovered pitch, and the starting position of these impulses may not be identical to the phase of the original speech. For unvoiced frames, random noise is used as the excitation, so the reconstructed waveform naturally changes each time even when the same parameters are used. Small timing differences may also appear because the signal is processed frame by frame, with windowing, overlap, and LPC filter memory between consecutive frames. However, these differences should not mainly be described as a communication synchronization failure. Since all frames were recovered and the payload BER was zero, the packet detection and symbol synchronization stages worked successfully for this experiment. The visible difference is therefore mainly caused by LPC-based compression, parameter quantization, excitation generation, and frame-based reconstruction. Also, the original signal in the upper graph was multiplied by eight only for visualization, so the amplitudes of the two signals should not be compared directly. The important result is that the reconstructed signal appears in the same main time regions and follows the general speech activity and energy changes of the original signal.



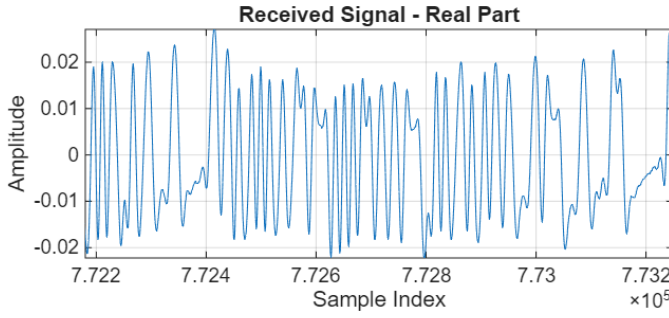


Fig. 7: Received signal baseband

Figure 7 shows a zoomed-in section of the real part of the received baseband signal. The waveform does not appear as a clean sequence of only +1 and -1 values, even though BPSK modulation was used. This is normal because the graph shows the received sample-level waveform before the final symbol decisions, not the original BPSK symbols themselves. Each symbol was represented by eight samples and passed through a Root Raised Cosine pulse-shaping filter. As a result, the transition between consecutive symbols becomes smooth and each symbol affects several nearby samples. The signal is also changed by the antennas, wireless channel, noise, reflections, attenuation, and USRP hardware. These effects cause the amplitude and waveform shape to vary over time.

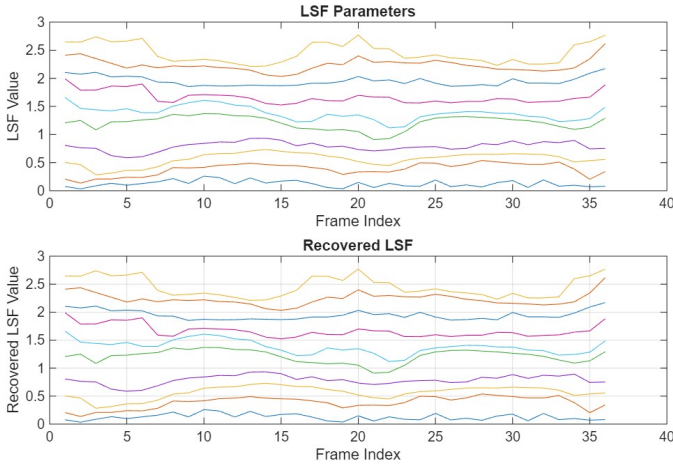


Fig. 8: Comparison of transmitted and recovered LSF parameters

Figure 8 compares the LSF parameters sent by the transmitter with the LSF parameters obtained at the receiver. The upper graph shows the ten LSF values before transmission, and the lower graph shows the same ten values after reception and decoding. Each colored line represents one LSF coefficient, and these coefficients describe the frequency shape of the speech in each frame. At the receiver, the LSF values are converted back into LPC coefficients and used to reconstruct the speech signal. When the two graphs are compared, the lines have the same shape and change in the same way from frame to frame. This means that the receiver recovered the transmitted LSF values correctly. The lines also remain in the

same order and do not cross each other, which is important because an incorrect order can make the LPC synthesis filter unstable. However, the graph compares the quantized LSF values at the transmitter and receiver. Before transmission, each LSF coefficient was converted into a 10-bit value, so some detail from the original coefficient was already removed during quantization. Therefore, Figure 8 shows that the wireless transmission did not change the LSF data, but it does not show that the reconstructed speech must be exactly the same as the original speech. The difference between the original and reconstructed speech can still come from quantization and the LPC synthesis process.

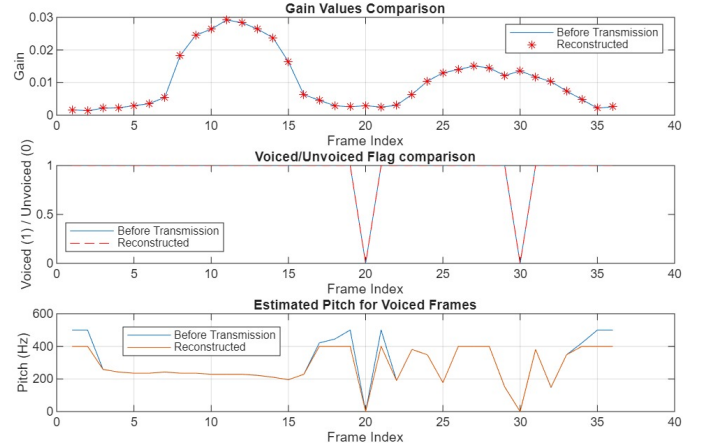


Fig. 9: Sent and recovered gain, voiced/unvoiced and pitch values

Figure 9 compares the gain, voiced/unvoiced decision, and pitch values before transmission and after reception. In the first graph, the transmitted and recovered gain curves almost overlap. Gain shows the energy level of each speech frame, so this result means that the loud and quiet parts of the speech were recovered correctly. In the second graph, the voiced/unvoiced values show whether a frame contains a periodic speech sound or a noise-like sound. A value of 1 represents a voiced frame, while a value of 0 represents an unvoiced frame. The transmitted and recovered values are almost the same, which means that the receiver selected the correct excitation type during speech reconstruction. In the third graph, the pitch values before and after transmission follow the same pattern. Pitch represents the period of the vocal-cord vibration and is mainly important for voiced frames. Sudden changes or zero values can appear near unvoiced frames because pitch is not meaningful in those regions. Overall, Figure 9 shows that the main speech parameters needed for reconstruction were recovered correctly. Therefore, the remaining difference between the original and reconstructed speech is not mainly caused by transmission errors, but by quantization and the LPC synthesis process.

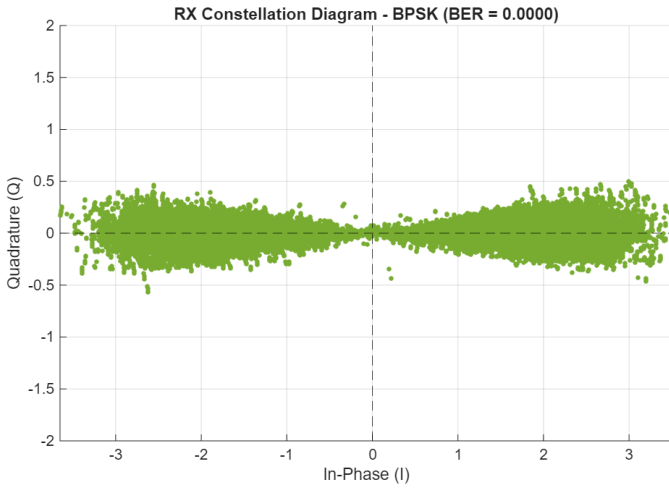


Fig. 10: Constellation diagram of the received signal

After the received signal was collected from the USRP, the DC component was removed and a matched filter based on the same Root Raised Cosine (RRC) filter used at the transmitter was applied. The filtered signal was then sampled using all possible symbol timing offsets, and the offset that produced the highest preamble correlation score was selected. The resulting symbol samples were normalized and a phase correction procedure was applied to compensate for residual phase offset. Finally, the recovered complex symbol samples corresponding to the detected packets were plotted on the I-Q plane to generate the RX constellation diagram.

The RX constellation diagram shows two clearly separated symbol clusters located on the positive and negative sides of the in-phase axis, which is the expected behavior for a BPSK system. The fact that the clusters remain well separated and centered around the real axis indicates that symbol detection was successful and that the phase correction stage operated correctly. Although the received symbols exhibit some spreading due to channel noise, filtering effects, timing imperfections, and hardware non-idealities, the separation between the two decision regions is maintained. This is consistent with the measured BER value of 0, indicating that all payload bits were recovered correctly despite the observed symbol dispersion.

Bit error rate was calculated by comparing the recovered payload bits with the originally transmitted payload bits after packet detection and frame reordering, and the result came out as 0 BER. This means that the bits are received with absolute correctness for this run.

From this receiving signal, after matched filtering, timing synchronization and phase correction, we perform BPSK hard decision detection to calculate the Bit Error Rate (BER). In this method, packet boundaries are determined based on the transmitted preamble sequence, and received payloads are rearranged based on frame IDs. This constructed bitstream was then compared, in a bit-by-bit fashion, to the original transmitted payload bitstream which encoded the

quantized LSF, gain, pitch and voiced/unvoiced parameters. BER was then calculated as incorrectly received bits divided by compared bits. The reached BER (bit error rate) was equal to 0.00000 meaning that all transmitted payload bits were recovered properly. Thus, this result shows that the communication system performed error-free and quantization is possible to any remaining difference in original and reconstructed speech signal caused by LPC vocoder approximation rather than transmission errors.

The observed SNR values were generally between -0.5 dB and -5 dB for our various system runs, which indicates that the reconstructed waveform is not very close to the original waveform in a sample-by-sample sense. Yet this does not mean that the wireless transmission did not work. The encoded speech parameters are correctly recovered from each bit of the information, as BER is measured 0.00000 for this case. Low/negative SNR is nearly completely due to the LPC vocoder structure: The system transmits LSF, gain, pitch, and voiced/unvoiced information in preference of original speech waveform. Thus, the receiver then synthesizes the speech from an impulse train or noise excitation that keeps general speech characteristics, not able to restore specific waveform of original signal. Consequently, the system performs well from the perspective of digital transmission; however, reconstruction quality suffers because LPC modeling, quantization and excitation are all approximated.

The reconstructed speech had a robotic sound, but it was understandable. The "MATLAB" speech was heard and registered by random listeners and the instructor. The causes of the robotic sound are mainly comes from LPC vocoder approximation, pitch/LSF quantization, and simplified excitation model. We chose 10 bits per LSF parameter to keep the message length short.

#### B. Data Size Comparison Before and After DSP-Based Representation

One of the main reasons for using LPC/LSF-based speech representation was to reduce the amount of data that had to be transmitted. If the prepared speech signal was transmitted directly as raw audio samples, the required bitrate would be much higher. In our system, the prepared speech signal had a sampling rate of 8 kHz. Assuming 16 bits per sample, the raw speech bitrate becomes

$$8000 \times 16 = 128000 \text{ bits/s} = 128 \text{ kbps.}$$

Instead of transmitting these raw samples, each speech frame was represented using speech parameters. In our implementation, each frame contained 10 LSF coefficients, one gain value, one pitch value, and one voiced/unvoiced flag. Each LSF coefficient was represented with 10 bits, gain was represented with 8 bits, pitch was represented with 8 bits, and the voiced/unvoiced flag was represented with 1 bit. Therefore, the payload size for one frame was

$$10 \times 10 + 8 + 8 + 1 = 117 \text{ bits/frame.}$$

The frame length was 160 samples at 8 kHz, which corresponds to 20 ms. With 25% overlap, the frame length becomes irrelevant to actual compression rate calculation and the newly defined hop length which is the difference between the frame length and its one-fourth, 120 samples. In the implemented data size calculation, this frame duration gives 66.7 frames per second. Therefore, the DSP-based speech parameter bitrate was calculated as

$$117 \times 66.7 = 7803.9 \text{ bits/s} = 7.8 \text{ kbps.}$$

The compression ratio between the raw speech bitrate and the DSP-based representation is

$$\frac{128}{7.8} \approx 16.41.$$

This result shows that the LPC/LSF-based representation greatly reduced the amount of data compared to sending the raw speech waveform. The comparison above only includes the speech parameter payload. Extra bits such as the preamble, frame ID, and repeated packets were added later for packet detection and wireless transmission, so they are not included in this payload bitrate comparison.

#### IV. CONCLUSION

In this project, a corrupted speech signal was processed, compressed, transmitted wirelessly, and reconstructed at the receiver. First, the signal was low-pass filtered and downsampled from 48 kHz to 8 kHz. Then, instead of transmitting all speech samples, the signal was represented using LSF, gain, pitch, and voiced/unvoiced parameters. This reduced the payload bitrate from 128 kbps to approximately 7.8 kbps, which corresponds to a compression ratio of about 16.41.

The generated bitstream was transmitted using BPSK modulation through the USRP and antennas. At the receiver, matched filtering, timing recovery, phase correction, packet detection, and hard-decision detection were applied. In the final experiment, all 36 frames were recovered and the payload BER was measured as zero. This means that no bit errors were observed in the compared payload bits during this experimental run. Although the received constellation was not ideal and showed spreading, the positive and negative BPSK symbols were still separated well enough for correct bit decisions.

The reconstructed speech was understandable, but it sounded robotic and its waveform was different from the original speech. This difference was mainly caused by the LPC vocoder, parameter quantization, and the simplified excitation signals used during reconstruction. Therefore, the low or negative waveform SNR does not mean that the wireless transmission failed. The transmission part recovered the speech parameters correctly, but the reconstruction method could not reproduce the exact waveform and natural sound of the original speech.

The system still has some limitations. It was tested under a limited number of gain, distance, and channel conditions, so the same BER result may not be obtained in a more difficult wireless environment. The speech quality could be improved by using proper overlap-add synthesis, a better excitation model, or a more advanced speech codec. Carrier and timing recovery could also be improved to obtain a cleaner constellation. In addition, CRC or FEC methods could be added to detect and correct errors. Overall, the project showed that speech parameters can be transmitted successfully with a much lower data rate by combining DSP methods, BPSK modulation, and USRP-based wireless communication.